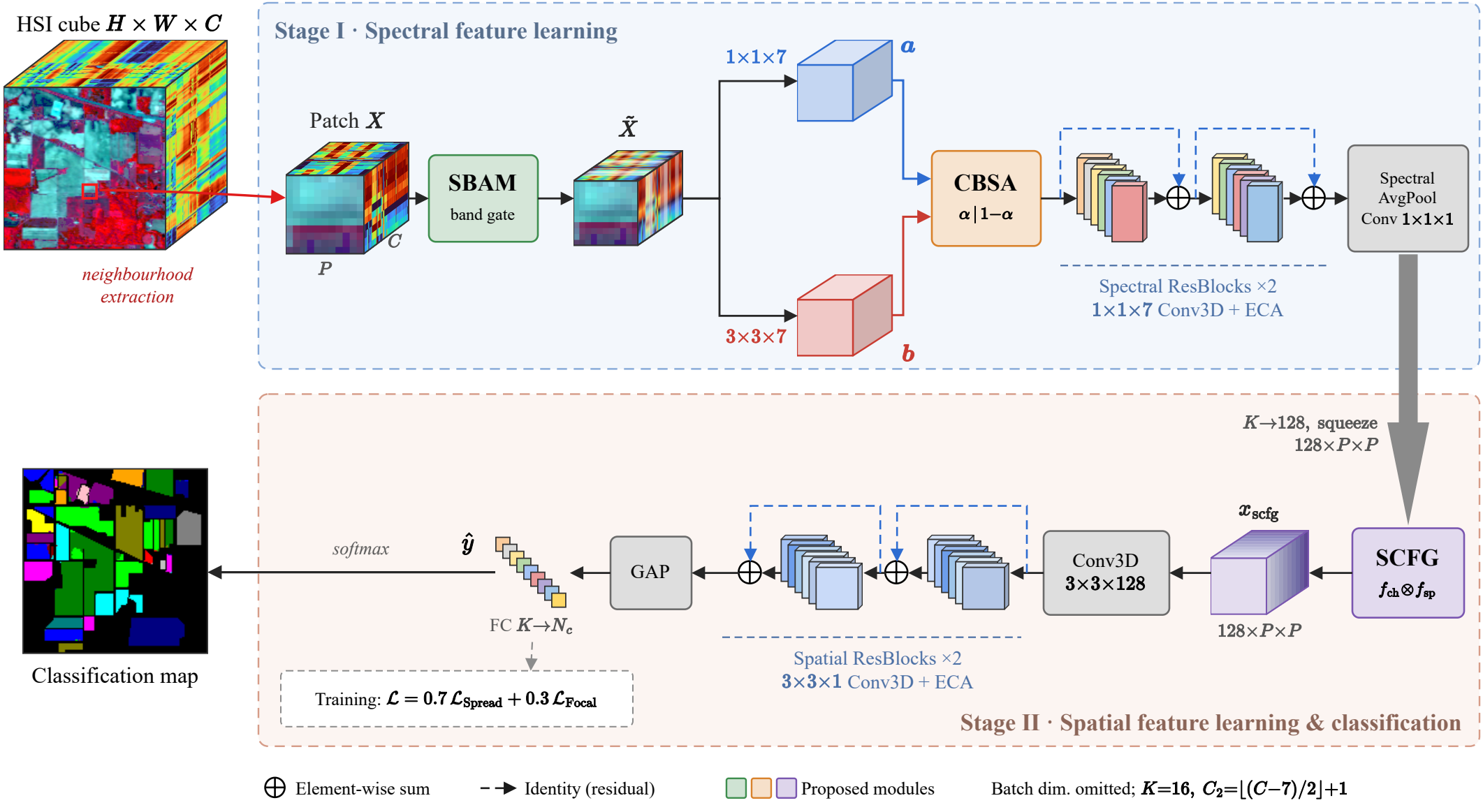

HSI cube $H \times W \times C$
neighbourhood extraction
Stage I · Spectral feature learning
Patch X
P
C
SBAM
band gate
$\tilde{X}$
1×1×7
a
3×3×7
b
CBSA
$\alpha \mid 1-\alpha$
Spectral ResBlocks ×2
1×1×7 Conv3D + ECA
Spectral AvgPool Conv 1×1×1
$K \to 128$, squeeze
$128 \times P \times P$
x_{scfg}
SCFG
$f_{ch} \otimes f_{sp}$
$128 \times P \times P$
Conv3D
3×3×128
Spatial ResBlocks ×2
3×3×1 Conv3D + ECA
GAP
FC $K \to N_c$
softmax
$\hat{y}$
Classification map
Training: $\mathcal{L} = 0.7\,\mathcal{L}_{Spread} + 0.3\,\mathcal{L}_{Focal}$
Stage II · Spatial feature learning & classification
Element-wise sum
Identity (residual)
Proposed modules
Batch dim. omitted; K=16, $C_2 = \lfloor (C-7)/2 \rfloor + 1$